Phase 0.0 — Slide Rule Basics

Type: Tool drill, no mission narrative. **Target time:** ~30 min if the scales are already comfortable; up to an afternoon if not. **Prerequisite:** None. This is the dependency root for everything else in the campaign. **Assumes:** A generic linear slide rule with C, D, A, B, K, S, and T scales (a standard 10" engineering rule — Pickett, K&E, Post, etc. all work). This is the baseline for Phase 0; nothing stops a later stage from calling for a specialized rule (vector rule, log-log duplex for exponentials, a period Trig/Kepler-specific rule, etc.) if the era or problem genuinely warrants it — just note the swap explicitly in that problem's file when it happens.

A note before you start

You've got E6B experience, not linear-rule experience — treat that as a **false friend**, not a head start. The E6B is a *circular* rule built around wind-triangle and fuel-burn conventions (rotating compass rose, specialized scales for TAS/groundspeed). A linear C/D/A/B rule shares the underlying logarithmic-scale principle but none of the E6B's specialized layout. Two habits from the E6B will actively work against you here:

- **No fixed "index" story.** On the E6B you're often reading off a wind triangle with a consistent visual frame. On a linear rule, the C and D scales slide relative to each other and there's no rotating dial to anchor your eye — you have to track which index (1 at the left end of C, or the 10 at the right end) you're using on a given problem, and that choice changes every time.
- **Decimal placement is entirely on you.** The E6B's problems are bounded (airspeeds, fuel flows) so wrong-order-of-magnitude answers are obviously wrong at a glance. A slide rule reads the same for 2.35, 23.5, and 235 — the scale only gives you the *significant digits*. You have to track the decimal point yourself with a quick mental estimate before you trust the rule's answer. This is arguably the single most important habit in this whole drill, and it's the one place hand computation fails silently if you skip it.

Scale reference

Scale	What it is	Use
C, D	Identical logarithmic scales, 1–10	Multiplication, division
A, B	Two decades (1–100) squeezed into the C/D length	Squares, square roots
K	Three decades (1–1000)	Cubes, cube roots
S	Sine scale, angle in degrees	sin(θ) read against C/D
T	Tangent scale, angle in degrees	tan(θ) read against C/D

Standard technique, if it's been a while: to multiply $a \times b$, set the left (or right) index of C over a on D, slide the cursor to b on C, read the result on D. Division reverses this. Squares: set cursor on D, read A. Cubes: set cursor on D, read K. Sine/tangent: set cursor to the angle on S (or T), read the corresponding value on C (aligned back to D for a combined calculation).

Drill 1 — Multiplication and division (C/D scale)

Work these on the C/D scale only. Estimate the decimal placement mentally *before* you read the rule, then confirm.

1. 2.35×4.12
2. 7.68×1.94
3. 3.14×6.02
4. $56.3 \div 8.15$
5. $128 \div 3.7$
6. 9.81×2.71828
7. 0.0456×273
8. $84.2 \div 0.0692$
9. $(4.5 \times 6.3) \div 2.1$ — do this as one continuous setting, without resetting the rule between the multiply and the divide. That chaining is the actual skill; a real hand-computer never re-set the rule for an intermediate result if they could avoid it.

Drill 2 — Squares, cubes, and roots (A/B, K scale)

1. 3.85^2
2. 12.7^2
3. 0.542^2
4. 2.15^3
5. $\sqrt{56.3}$
6. $\sqrt{842}$
7. $\sqrt[3]{47.5}$

Drill 3 — Trig scales (S, T)

1. $\sin(23.5°)$
2. $\sin(67°)$
3. $\tan(15°)$
4. $\tan(52°)$
5. $45.0 \times \sin(30.0°)$ — combined trig + multiply, one setting
6. $68.3 \div \tan(41.0°)$ — combined trig + divide, one setting

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Worksheet

Fill in your slide rule reading and your decimal-placement estimate *before* checking the answer key.

Problem	Your estimate (order of magnitude)	Your slide rule reading
1.1		
1.2		
1.3		

Problem	Your estimate (order of magnitude)	Your slide rule reading
1.4		
1.5		
1.6		
1.7		
1.8		
1.9		
2.1		
2.2		
2.3		
2.4		
2.5		
2.6		
2.7		
3.1		
3.2		
3.3		
3.4		
3.5		
3.6		